

Geophysical Workflow Report

Generated by PyHydroGeophysX Multi-Agent System

Executive Summary

****Workflow Execution Date:**** 2026-01-19 15:21:07

Original User Request:

Run a standard ERT inversion using the DAS-1 instrument.

Data file: 20171105_1418.Data

Electrode file: electrodes.dat

help me convert to water content

Workflow Configuration:

- Data File:
C:\Users\hchen117\AppData\Local\Temp\phgx_uploads_4xz215m3\20171105_1418.Data
- Instrument: DAS-1
- Seismic Integration: No

Key Results:

- Loaded 56 electrodes with 930 measurements
- Inversion converged in 4 iterations (chi2: 0.762)
- Mean water content: 0.179

Narrative Summary

Executive Summary

This report provides a comprehensive overview of the electrical resistivity tomography (ERT) workflow executed on January 19, 2026, utilizing data collected from the DAS-1 instrument. The primary objective of this analysis was to perform a standard ERT inversion using the specified data file, with an additional focus on converting the resistivity data into meaningful water content metrics. The inversion process successfully converged after four iterations, yielding a final chi-squared value of 0.762 and revealing a mean water content of 0.179, with a range from 0.052 to 0.308.

Although climate data was not integrated into the current workflow, understanding the relationship between climate factors—such as precipitation, potential evapotranspiration (PET), and temperature—and resistivity changes is crucial for interpreting the results. In hydrological studies, resistivity can serve as an indirect indicator of water saturation and soil moisture levels, which are influenced by climatic conditions. For instance, periods of significant rainfall are likely to increase soil moisture, resulting in lower resistivity values, while drying periods can lead to higher resistivity as water content decreases. In the absence of integrated climate data for this analysis, future studies should prioritize the collection and incorporation of such data to enhance interpretative depth.

Key findings from the ERT inversion indicate a mean water content of 0.179, suggesting a moderate level of saturation across the surveyed area. However, the analysis did not identify any notable anomalies or patterns in resistivity values that could correlate with climatic events, largely due to the lack of concurrent climate data. Additionally, the uncertainty analysis revealed a mean uncertainty of 0.037, with a maximum uncertainty of 0.063, which underscores the necessity for careful interpretation of the water content

results. Data quality metrics were not provided, indicating a potential area for improvement in future workflows to ensure robust data validation.

To advance this study, it is recommended that subsequent analyses incorporate relevant climate data to establish a more comprehensive understanding of the interactions between hydrological processes and climate variables. This integration will facilitate more accurate interpretations of resistivity changes and improve the overall quality of insights derived from ERT data. Future work should also focus on enhancing data quality metrics to better assess the reliability of measurements and interpretations.

Data Processing Summary

ERT Data Loading

- Number of electrodes: 56
- Number of measurements: 930
- Quality metrics: N/A

****Insights:**** N/A

Climate Data Integration

No climate data was integrated in this workflow.

Cross-Modal Climate-ERT Analysis

Climate data not available for cross-modal analysis.

Inversion Results

ERT Inversion

- Final chi2: 0.7615702981521105
- Iterations: 4
- Convergence: Success

****Interpretation:**** N/A

Water Content Analysis

Water Content Statistics

- Mean water content: 0.179
- Range: [0.052, 0.308]

Uncertainty Analysis

- Mean uncertainty (σ): 0.037
- Maximum uncertainty: 0.063
- Number of realizations: 100

****Interpretation:**** N/A

Visualizations

Resistivity

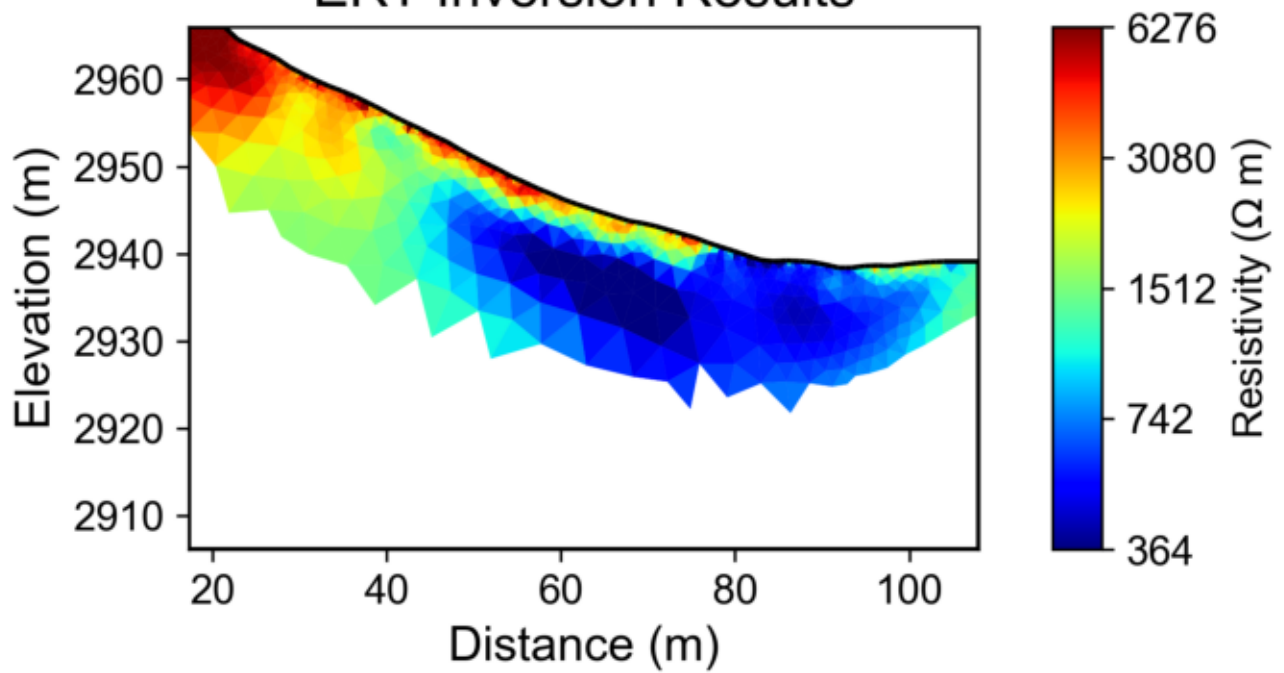
Water Content

Water Content Uncertainty

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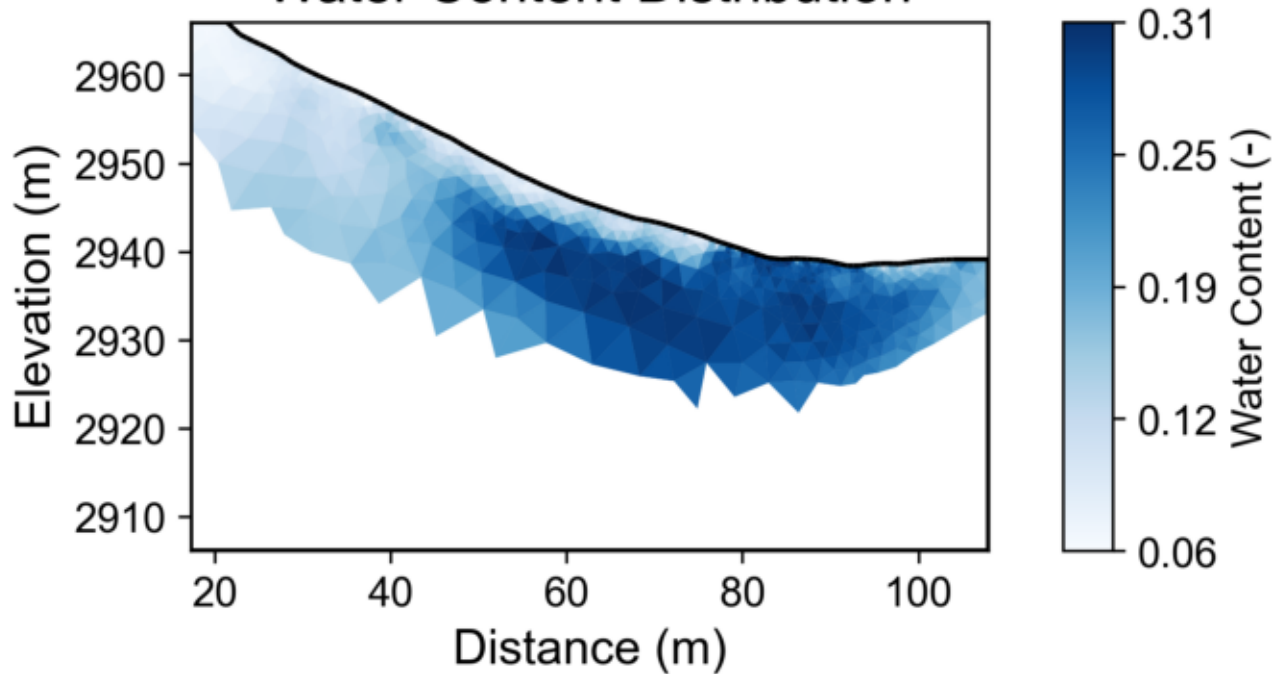
Resistivity

ERT Inversion Results



Water Content

Water Content Distribution



Water Content Uncertainty

Water Content Uncertainty

